

Math 110 – Abstract Linear Algebra

Chapter-by-chapter notes on the important ideas

Axler, *Linear Algebra Done Right* (4th ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Taking notes

Find the chapter banner, then type under the section heading as you read:

`\notesec{1A}{Span and Linear Independence}`

Record big ideas with `idea`, definitions with `definition`, theorems with `theorem`, `lemma`, `proposition`, examples with `example`, and warnings with `remark` — all numbered within the chapter. Add a missing section by copying a `\notesec` slot; add a chapter with `\chapterbanner`.

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Chapter 1 Vector Spaces

Axler: $\mathbb{F}^n$, vector space axioms, subspaces, sums and direct sums

1A $\mathbb{R}^n$ and $\mathbb{C}^n$ **1B Definition of Vector Space****1C Subspaces**

Chapter 2 Finite-Dimensional Vector Spaces

Axler: span, linear independence, bases, dimension

2A Span and Linear Independence**2B Bases****2C Dimension**

Chapter 3 Linear Maps

Axler: linear maps, null space and range, matrices, invertibility and isomorphisms, products and quotients, duality

3A Vector Space of Linear Maps

3B Null Spaces and Ranges

3C Matrices

3D Invertibility and Isomorphisms

3E Products and Quotients of Vector Spaces

3F Duality

Chapter 4 Polynomials

Axler: zeros, the division algorithm, factorization over $\mathbb{C}$ and over $\mathbb{R}$

4.1 Zeros of Polynomials

4.2 Division Algorithm for Polynomials

4.3 Factorization of Polynomials over $\mathbb{C}$

4.4 Factorization of Polynomials over $\mathbb{R}$

Chapter 5 Eigenvalues and Eigenvectors

Axler: invariant subspaces, eigenvalues, the minimal polynomial, upper-triangular matrices, diagonalizability, commuting operators

5A Invariant Subspaces

5B The Minimal Polynomial

5C Upper-Triangular Matrices

5D Diagonalizable Operators

5E Commuting Operators

Chapter 6 Inner Product Spaces

Axler: inner products and norms, orthonormal bases and Gram–Schmidt, orthogonal complements, minimization

6A Inner Products and Norms

6B Orthonormal Bases

6C Orthogonal Complements and Minimization Problems

Chapter 7 Operators on Inner Product Spaces

Axler: adjoints, self-adjoint and normal operators, the spectral theorems, positive operators, isometries, SVD

7A Self-Adjoint and Normal Operators**7B The Spectral Theorem****7C Positive Operators****7D Isometries, Unitary Operators, and Matrix Factorization****7E Singular Value Decomposition****7F Consequences of Singular Value Decomposition**

Chapter 8 Operators on Complex Vector Spaces

Axler: generalized eigenvectors, nilpotent operators, generalized eigenspace decomposition, Jordan form, trace

8A Generalized Eigenvectors and Nilpotent Operators

8B Generalized Eigenspace Decomposition

8C Consequences of Generalized Eigenspace Decomposition

8D Trace: A Connection Between Matrices and Operators

Chapter 9 Multilinear Algebra and Determinants

Axler: bilinear and quadratic forms, alternating multilinear forms, determinants, tensor products

9A Bilinear Forms and Quadratic Forms**9B Alternating Multilinear Forms****9C Determinants****9D Tensor Products**